

Randomness Extraction from Unpredictable Random-Oracle Sources

A Leftover-Hash-Lemma Conjecture for the Secret-Seed Game

Status. Statement: human-written, not yet formalized. Proof: open – no attempt yet.

Category. Information-Theoretic Cryptography.

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Conjecture (informal). *For an extractor built from a random function with a K -element seed space, and any source whose output no unbounded predictor can guess with probability better than ϵ – even given the extractor’s entire function table – the extracted value should look uniform to an unbounded distinguisher that does not see the seed, up to statistical distance about $\sqrt{(\epsilon R + \log D)/K}$. This is the shape of the classical leftover hash lemma, but with unpredictability against an omniscient adversary in place of min-entropy as the source restriction, and a single universal constant working for every choice of parameters.*

1 The setting

An extractor built from a random function H maps a seed and an unpredictable source’s output to a value that should look uniform to an unbounded distinguisher holding the source’s auxiliary output and the oracle’s entire table. Unpredictability is measured against an equally unbounded predictor with the same information. Whether the distinguisher also learns the seed splits the question into two games; this note concerns the harder one, where the seed stays hidden from it. Definition 1 formalizes both notions.

Notation. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. Let $\text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$

be the set of all functions $\mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}$, let $\text{SD}(\cdot, \cdot)$ denote statistical distance and $U_{\mathcal{R}}$ the uniform distribution on $\mathcal{R}$. The source S , the predictor P and the distinguisher D of Figure 1 are computationally unbounded and receive the *entire* function table of H as an explicit input.

2 The conjecture

Notation as in Section 1. Definition 1 restates the games formally, for self-containment; the conjecture quantifies over all ϵ -unpredictable sources.

Game Pred_P^S $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\\$} S(H); \quad x' \xleftarrow{\\$} P(H, z);$ return $(x = x')$
Game Ext_D^S (secret seed) $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\\$} S(H); \quad sd \xleftarrow{\\$} \mathcal{K}$ $y_0 \leftarrow H(sd, x); \quad y_1 \xleftarrow{\\$} \mathcal{R}; \quad b \xleftarrow{\\$} \{0, 1\}; \quad b' \xleftarrow{\\$} D(H, y_b, z);$ return $(b = b')$

Figure 1: Prediction game (top) and the secret-seed extraction game (bottom); unlike the public-seed variant (Remark 1), D does not receive sd .

Definition 1 (the extraction games). Set $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) := \Pr[\text{Pred}_P^S \Rightarrow 1]$ and

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) := 2 \Pr[\text{Ext}_D^S \Rightarrow 1] - 1.$$

A source S is ϵ -unpredictable if $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) \leq \epsilon$ for every unbounded predictor P .

Conjecture 1 (secret seed). *There is a universal constant $c > 0$, independent of $\mathcal{K}$, $\mathcal{D}$, $\mathcal{R}$ and ϵ , such that for all nonempty finite*

$\mathcal{K}, \mathcal{D}, \mathcal{R}$, all $\epsilon \in (0, 1]$, every ϵ -unpredictable source S and every unbounded distinguisher D ,

$$\mathbf{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) \leq \delta(\epsilon, K, D, R) := c \cdot \sqrt{\frac{\epsilon R + \log_2 D}{K}}.$$

Remark 1 (the public-seed variant is already resolved). Giving the seed to D as well defines a second game, Ext-pub, with advantage $\mathbf{Adv}^{\text{ext-pub}}$. Unlike the conjecture above, its natural bound is proved: for the same universal constant $c = 2$,

$$\mathbf{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) \leq c \cdot \left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}} \right)$$

holds unconditionally (see <https://crypto-conjura.github.io/c/0004/>). This is weaker than Conjecture 1 whenever the two are compared directly, since $\sqrt{(\epsilon R + \log_2 D)/K} \leq \sqrt{\epsilon R} + \sqrt{\log_2 D/K}$ – consistent with hiding the seed only helping the extractor, never hurting it.